

DRY-RUN GUIDE | 18_isomorphic_strings.py
Source: Python\dsa_solutions\18_isomorphic_strings.py

CORE BEHAVIOR

Checks for a consistent one-to-one character mapping between strings.

REPRESENTATIVE INPUT

'egg', 'add'.

STEP TRACE

Map e->a, g->d; the repeated g maps to the same d.

EXPECTED OUTPUT / RETURN

True

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\dsa_solutions\18_isomorphic_strings.py; line numbers refer to the original file. No source was modified.

```
001 | import sys
002 |
003 |
004 | def are_isomorphic(first, second):
005 |     if len(first) != len(second):
006 |         return False
007 |     forward = {}
008 |     backward = {}
009 |     for left, right in zip(first, second):
010 |         if forward.get(left, right) != right or backward.get(right, left) != left:
011 |             return False
012 |         forward[left] = right
013 |         backward[right] = left
014 |     return True
015 |
016 |
017 | def main():
018 |     first, second = sys.stdin.read().split()
019 |     print("true" if are_isomorphic(first, second) else "false")
020 |
021 |
022 | if __name__ == "__main__":
023 |     main()
024 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.